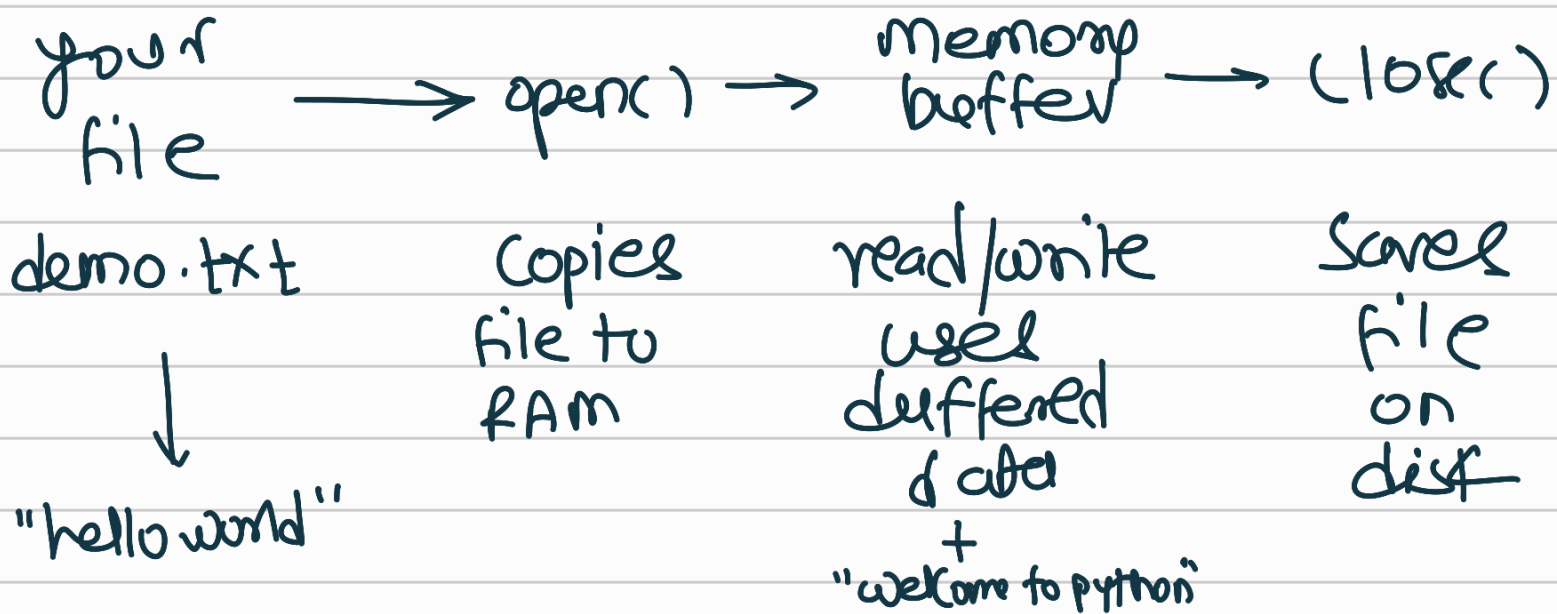


File Input - output in Python -

Text file — demo.txt

- ✓ new file creation
- ✓ file read — display file output
- ✓ file write — overwrite
- ✓ file write — append

How file I/O works -



RAM - Random Access memory.

ROM - Read only memory
(Disk space)

HDD - 500GB

SDD - 256GB

folder 2
└─ demo.txt

eg. demo.txt — "hello world"

folder 1

└─ file1.py
└─ demo.txt

"welcome to python"

fobj = open("demo.txt", "r")

file obj file name/path mode

'r'
"w"
'a'

result = fobj.read()

print(result)

fobj.close()

Save file
on disk

Output
display

RAM memory
Read operation

file
loaded
into
RAM
①

2] write operation:-

msg = """ Hello world

welcome to AI Engineer Course """

```
fobj2 = open("file1.txt", "w")
```

```
fobj2.write(msg)
```

```
fobj2.close()
```

- without "with" - risky

```
fobj = open("demo.txt", "r")  
→ result = fobj.read()  
  print(result)  
→ fobj.close()
```

Diagram annotations: An arrow points from the 'r' in the first line to a circled '1'. An arrow points from the 'read()' method to a circled '2'. An arrow points from the 'close()' method to a circled '3'.

① if exception occurs before close()

② file may get locked, blocking for other programs.

③ Data may not be fully written to disk.

Using with keyword (Safe & Simple, Clean)

without with? -

```
fobj = open("demo.txt", "r")
```

```
→ result = fobj.read()
   print(result)
→ fobj.close()
```

① ↓
② ↓
③ ↓

using with :-

```
① → with open("demo.txt", "r") as fobj :
    ② →     result = fobj.read()
    ③ →     print(result)
```

① file closes automatically when block exits.

② No need to call close() manually

③ Simple readable code

④ works even with exception raised.